

Here is a **comprehensive, step-by-step checklist** for performing a **SQL Server in-place upgrade from 2012 to 2019 Enterprise Edition** in an environment configured with **Always On Availability Groups (AGs)**.

This guide includes:

- A **rolling upgrade strategy**
- A detailed **rollback plan**
- Key **pre- and post-upgrade validations**

✅ SQL Server 2012 to 2019 In-Place Upgrade Checklist in Always On Availability Group Environment

◆ ASSUMPTIONS

- You're running **SQL Server 2012 Enterprise** with **Always On AGs** enabled.
- Databases are synchronized and healthy across **multiple replicas**.
- You intend to do an **in-place upgrade** to **SQL Server 2019 Enterprise**, using the **rolling upgrade method** (node by node).

📌 1. PRE-UPGRADE PREPARATION

✅ 1.1 Infrastructure & Compatibility

- Confirm **SQL Server 2012 → 2019** upgrade is **supported** in-place for Always On AG ([Microsoft documentation](#)).
- Confirm the **Windows Server OS version** is compatible with SQL Server 2019.
- Ensure AGs are in **synchronous mode** and replicas are **healthy**.
- Validate cluster quorum configuration and node votes.
- Ensure **service packs and CUs** are up to date on SQL Server 2012 before upgrade.

✅ 1.2 Tools & Warnings

- Run **Data Migration Assistant (DMA)** to detect deprecated features.
- Run DBCC CHECKDB on all user databases.
- Review any **AG listener configurations**, endpoints, custom ports, certificates.
- Take note: once a **primary database is upgraded to SQL 2019**, you **cannot fail back** to 2012.

✅ 1.3 Backups & Snapshots

- Take **FULL backups** of:
 - All user databases
 - System databases (master, model, msdb)
- Export:
 - SQL Agent Jobs
 - Linked servers
 - Logins, credentials
 - AG configuration (via SSMS or PowerShell)
- Capture **VM snapshots or OS-level backups** of each replica (mandatory for rollback).
- If applicable, use **log shipping or backup-restore to create a cold standby** SQL 2012 environment for DR rollback.

2. ROLLING UPGRADE PROCESS

 The core idea: Upgrade **secondary replicas first**, fail over, then upgrade the former primary.

STEP 1: UPGRADE SECONDARY REPLICAS

For each **secondary replica** (one at a time):

- Suspend **automatic failover** and **data movement** (ALTER AVAILABILITY GROUP ... SUSPEND DATA MOVEMENT).
- Remove the replica from the AG (**optional but cleaner**).
- Stop applications from connecting (if used for read workloads).
- Perform **in-place upgrade** of SQL Server 2012 to 2019.
- Reboot the server after the upgrade.
- Rejoin the node to the AG (if removed).
- Resume data movement and synchronization.
- Validate synchronization state: sys.dm_hadr_database_replica_states.

 Repeat for all secondary replicas.

STEP 2: FAILOVER TO UPGRADED SECONDARY

- Choose one of the upgraded secondaries as the new **primary**.
- Perform a **manual failover** (ALTER AVAILABILITY GROUP ... FAILOVER or via SSMS).
- Validate AG health, synchronization, and listener configuration on new primary.
- Note: The databases on this replica are now **upgraded to SQL Server 2019 format** → cannot go back to 2012.

STEP 3: UPGRADE FORMER PRIMARY

- On the old primary (now secondary, still running SQL 2012):
 - Remove from AG (optional).
 - Upgrade SQL Server to 2019.
 - Reboot the server.
 - Rejoin AG and resume data movement.
- Validate synchronization.

3. POST-UPGRADE TASKS

- Update **database compatibility level** (e.g., to 150) after testing.
- Run DBCC CHECKDB on all databases.
- Update **statistics and rebuild indexes**.
- Review and re-enable:
 - Agent jobs
 - Maintenance plans
 - Monitoring tools
- Validate application performance and query plans.
- Test **automatic and manual failover** across upgraded nodes.

4. ROLLBACK PLAN – IN DETAIL

Rollback depends on **when failure occurs**:

SCENARIO 1: Rollback BEFORE Primary is Upgraded

If **only secondary replicas** are upgraded:

-  You can:
 - Detach/remove them from AG
 - Uninstall SQL Server 2019
 - Restore VM snapshot or reinstall SQL 2012
 - Rejoin AG after restoring

This is a **safe rollback point**. Always confirm app functionality before promoting a 2019 replica.

SCENARIO 2: Rollback AFTER Primary is Upgraded

If the **primary is already upgraded to SQL Server 2019**, databases are upgraded and cannot be attached to SQL 2012 again.

In This Case:

- Restore pre-upgrade full backups to a **clean SQL Server 2012 instance** (new or cold standby).
- Reconfigure Always On:
 - Create AG from scratch on 2012
 - Join other 2012 servers (also restored or rebuilt)
- Point applications to new environment.

Requirements:

- You **must** have full backups of all DBs, jobs, configs, and a **working cold standby**.
- Plan for **longer downtime** if rollback occurs at this stage.

Best Practice for Rollback Safety:

Time	Can Roll Back?	Method
Before any upgrade	 Yes	Just don't proceed
After secondary(s) upgraded	 Yes	Uninstall 2019, revert snapshot
After primary upgraded	 No	Must restore to SQL 2012 from backups

5. OPTIONAL: AUTOMATION AND SCRIPTING

- Use PowerShell to automate:
 - Backup of AG configuration
 - AG failover
 - Health checks
- Script the entire upgrade and validation workflow.
- Enable **Query Store** (SQL 2019 feature) for performance tracking.

 FINAL CHECKLIST SUMMARY

Task Area	Action Item
Pre-checks	Backups, snapshots, validations, DMA
Upgrade	Secondary → failover → primary
Post-checks	AG sync, jobs, performance, DB checks
Rollback Plan	Snapshot, cold standby, restore backups
Final Validation	Failover tests, listener tests, apps

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